

SITKA, AK, USA

WMO: 703710

Lat: 57.048N

Lon: 135.365W

Elev: 4

StdP: 101.27

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25333

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.9	-6.0	-18.8	0.7	-6.0	-16.0	0.9	-3.9	12.7	6.2	11.5	5.9	2.9	90	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.5	19.3	14.7	17.9	14.2	16.8	13.7	15.7	18.0	15.0	17.0	14.4	16.1	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	10.3	16.4	13.9	9.9	15.7	13.4	9.6	15.2	43.7	18.2	41.6	17.0	40.1	16.1	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.4	8.4	DB -9.9	23.3	2.3	2.4	-11.6	25.0	-12.9	26.4	-14.2	27.8	-15.9	29.5		
(5)			WB -11.3	17.1	2.5	1.9	-13.1	18.5	-14.5	19.6	-15.9	20.6	-17.6	22.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	2.7	2.6	3.2	5.9	8.9	11.7	13.6	14.1	11.9	8.1	4.5	3.0		
	DBStd	5.02	3.84	3.24	3.08	2.27	2.24	1.87	1.55	1.63	1.87	2.35	3.22	3.43		
	HDD10.0	1281	227	208	213	126	48	4	0	0	5	66	166	218		
	HDD18.3	3948	486	442	471	373	293	201	148	132	194	317	415	477		
	CDD10.0	376	1	0	1	2	14	53	110	127	61	7	1	0		
	CDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0		
	CDH23.3	4	0	0	0	0	0	1	1	2	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	3.4	3.9	3.6	3.7	3.4	3.1	2.8	2.6	2.6	3.0	3.8	4.1	4.3		
(15) Precipitation	PrecAvg	2218	207	160	153	125	117	93	104	169	286	352	236	214		
	PrecMax	3123	400	395	327	284	255	206	242	496	527	783	426	384		
	PrecMin	1527	41	6	22	29	31	23	42	44	100	161	42	57		
	PrecStd	374	90	92	66	60	59	50	45	99	117	128	94	87		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	11.3	12.3	16.4	20.0	20.7	20.2	21.8	19.8	14.8	12.5	11.2		
		MCWB	8.1	6.3	7.2	9.6	10.6	14.9	15.8	16.5	14.4	11.6	8.4	7.4		
	2%	DB	9.2	9.0	9.7	12.4	15.5	17.4	18.2	19.2	17.2	13.2	11.0	9.7		
		MCWB	6.5	5.6	6.1	7.8	10.0	13.5	14.8	15.3	13.0	10.8	8.0	6.9		
	5%	DB	8.2	7.8	8.3	10.5	13.6	15.9	17.0	17.9	15.9	12.3	9.9	8.4		
		MCWB	5.8	5.2	5.4	6.7	9.9	12.7	14.1	14.8	12.9	10.0	7.6	6.2		
	10%	DB	7.3	7.0	7.2	9.3	12.3	14.6	16.1	16.9	14.8	11.4	8.8	7.4		
		MCWB	5.3	4.8	4.9	6.2	9.4	11.9	13.7	14.3	12.6	9.3	6.9	5.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	7.6	8.1	10.3	12.2	15.5	16.3	17.1	15.5	12.8	10.1	8.8		
		MCDB	10.8	9.1	10.7	14.9	16.2	19.1	19.3	19.8	17.3	13.6	11.1	9.9		
	2%	WB	7.1	6.4	7.0	8.4	11.1	14.1	15.2	16.0	14.4	11.7	8.8	7.6		
		MCDB	8.4	8.0	8.8	11.2	14.1	16.6	17.6	18.3	15.9	12.5	10.1	8.9		
	5%	WB	6.3	5.7	6.1	7.6	10.5	13.2	14.6	15.2	13.7	10.7	7.9	6.6		
		MCDB	7.6	7.3	7.7	9.8	13.0	15.4	16.5	17.3	15.1	11.6	9.4	8.2		
	10%	WB	5.6	5.0	5.2	6.8	9.8	12.3	14.0	14.7	13.0	10.0	7.0	5.6		
		MCDB	7.0	6.7	6.9	8.9	11.9	14.0	15.6	16.4	14.4	11.1	8.6	7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.6	5.0	5.5	5.1	4.6	3.9	4.5	4.7	4.5	4.2	4.0		
		MCDBR	4.2	4.8	6.2	7.6	8.1	7.1	5.8	6.8	6.8	5.0	4.4	4.2		
	5% WB	MCWBR	2.7	2.9	3.9	4.4	4.5	3.9	3.4	3.9	4.1	3.5	3.2	3.0		
		MCWBR	3.8	4.3	5.4	6.9	6.8	6.3	5.1	5.8	5.2	3.9	4.1	4.1		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.313	0.332	0.396	0.374	0.371	0.382	0.362	0.356	0.340	0.305	0.282		
		taud	2.249	2.386	2.374	2.186	2.380	2.438	2.445	2.524	2.496	2.394	2.271	2.190		
	Ebn at Noon		568	746	826	811	863	870	851	848	797	697	554	475		
		Edh at Noon	51	68	88	124	109	105	102	89	79	67	49	41		
(44) All-Sky Solar Radiation	RadAvg		0.38	1.06	2.17	3.64	4.62	4.74	4.01	3.46	2.23	1.18	0.51	0.27		
	RadStd		0.06	0.21	0.31	0.35	0.49	0.43	0.47	0.44	0.37	0.16	0.07	0.04		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+1.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	+1.11	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page